

HodTech

Computing



HodTech Catalogue
Spring 2021

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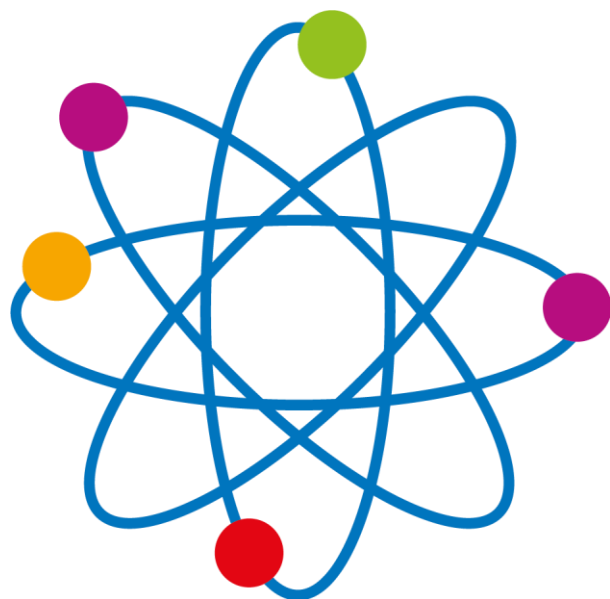
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Computing Components



RAM (Random Access Memory)



CPU (Central Processing Unit)



Motherboard



Power supply



Graphics card



Hard drive

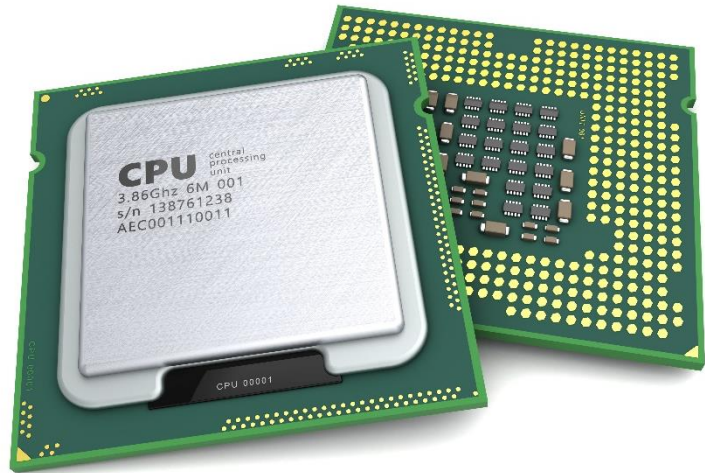


Optical drive

CPU

CPU stands for Central Processing Unit and the CPU carries out all the processing in a computer by computing the fetch–decode–execute cycle.

The processing carried out by the CPU allows you to do all the things you do on a computer, including arithmetic, sorting, searching and browsing the internet.



There are lots of different parts to the CPU and it has some memory that temporarily stores the instructions it needs to carry out processing.

The speed of a CPU is measured in hertz (Hz). A one-hertz processor can complete one fetch–decode–execute cycle per second. A one-kilohertz (KHz) processor can complete the fetch–decode–execute cycle 1000 times per second. And one-megahertz (MHz) processor can complete the fetch–decode–execute cycle 1 million times per second.

Modern CPUs run in gigahertz (GHz). A one gigahertz CPU can complete the fetch–decode–execute cycle 1 billion times per second.

Modern CPUs have more than one core, and you will come across terms like dual core and quad core, to refer to two and four cores respectively. If a CPU has more than one core, it can process more than one fetch–decode–execute cycle at the same time.

In theory, a dual-core CPU should be twice as quick as a single-core CPU, but this is not always the case. This is because some fetch–decode–execute cycles cannot be processed until other fetch–decode–execute cycles having been completed, so it isn't always possible to process two fetch–decode–execute cycles at the same time.

RAM

RAM stands for Random Access Memory.

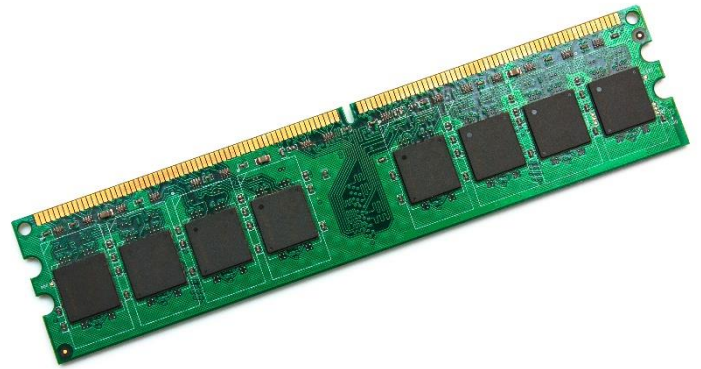
RAM is storage that is connected directly to the motherboard. RAM can help improve performance because data can be stored in the RAM can be accessed more quickly by the CPU than data stored on the hard drive.

The problem with RAM is that when you turn the computer off, it loses all the data it is storing. It is not permanent storage.

RAM is used to store the following:

- programs that are currently in use
- data that is currently being used
- parts of the operating system currently being used by the CPU.

Most modern computers have 4–8 GB of RAM for temporary storage. The speed of RAM is also measured, usually in megahertz (MHz), and modern RAM can run at over 3000 MHz.

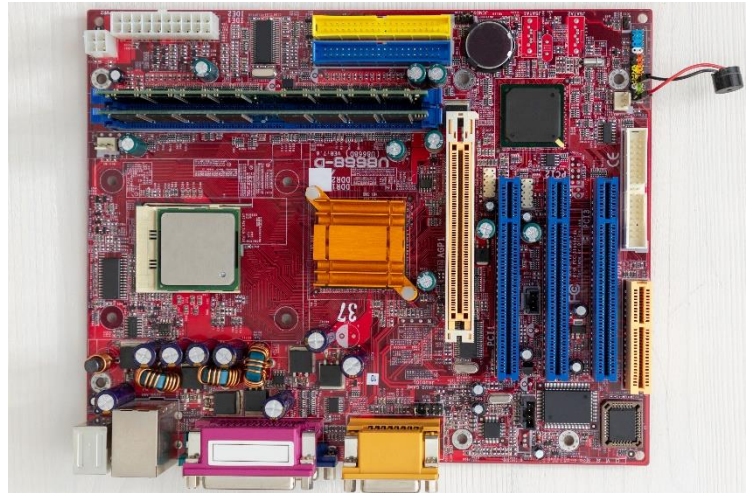


Mother-board

The motherboard is the main circuit board found in a computer. It is the component that connects all the components together, so they can communicate.

Some components connect directly to the motherboard. These include the CPU, the graphics card and the RAM.

Other components connect to the motherboard using cables. These include the hard drive and optical drive.



The motherboard also contains power connectors. The power supply connects to the motherboard and the motherboard sends it on to some of the components that connect directly to it.

Motherboards can come in different sizes, so they can fit in devices of different sizes. The size of a motherboard is known as its 'form factor'. The smaller the form factor, the fewer expansion slots you usually have to plug in additional graphics cards or sound cards.

Modern motherboards usually have integrated graphics cards, sound cards and network cards, so there is less need to purchase cards to plug in. However, the integrated cards may be basic and you may want to buy a dedicated graphics card if you want to run powerful games.

Graphics card

A graphics card is an additional component that plugs directly into the motherboard. Some motherboards have integrated graphics cards, but they are often far less powerful than a dedicated graphics card.



Dedicated graphics cards connect into either the AGP or the PCI express connection port on the motherboard.

Graphics cards are used to output and display graphics on screen. Graphics cards work in a similar way to the CPU but they have a dedicated function: they process graphics. They have their own processor, called a GPU (Graphics Processing Unit).

A graphics card often has its own cooling mechanism, just like a CPU. This is because it gets hot when it is processing. How much cooling a graphics card needs depends on how powerful it is; the more processing it does, the more cooling it needs. Having its own cooling mechanism means that a graphics card needs its own power supply and it has power connectors that allow the power supply to plug directly into it.

A graphics card often has many different connectors, including VGA, DVI and HDMI, to allow you to plug different types of monitor into it.

The most demanding games and applications may require more than one graphics card. Some graphics cards can be linked together to spread the processing, in a similar way to a dual-core CPU.

Power supply

The power supply provides the power to all the components in a computer that need it.

Power supplies for a standard computer all follow the same standard, known as ATX. This means that all the connectors used are standardised. This helps companies making other components know what power connectors to use.



The more powerful the component in a computer, the more power it needs from the power supply. Some modern graphics cards use 300 watts of power on their own!

If you do not provide the components in a computer with enough power, you might find that your computer freezes or does not boot up in the first place.

Hard drive

A hard drive, or hard disc, is the permanent storage in a computer. There are two main types of hard drive: magnetic hard drives and Solid State Drives.

A magnetic disc has a stack of magnetic platters that rotate. Moving read/write heads hover just above the surfaces and can move in and out over the disc surfaces to read or write the data.



A modern magnetic hard drive can have over 1 terabyte of storage space, although you may not have access to all of this storage because some of it is used by the operating system. The speed of a hard drive is measured by how long it takes to find data and also by how fast the discs spin, measured in RPM.

Solid State Drives (SSDs) are relatively new. As yet, they do not have the same capacity as the traditional magnetic hard drive in an average home computer. Most SSDs have only 128 GB or 256 GB of storage space. Solid State Drives have no moving parts and can, therefore, find data extremely fast.

Optical drive

An optical drive uses a laser to read or write to optical discs. Some optical drives can only read certain types of discs, such as DVDs, while others can read all types of discs, including CDs, DVDs and Blu-ray Discs™.

Some optical drives allow you to 'burn' or write to optical discs. This means you can record data onto a blank disc. This was extremely popular before streaming services became widespread, as people used to copy CDs and DVDs.



Optical drives come in different sizes, depending on the size of the device they are designed to fit into. However, the diameter of all optical drives is the same, because all optical discs are the same size.

Optical drives are found not just in computers. They can also be found in gaming consoles and stereo systems, as well as many other devices.

Storage



Magnetic



Optical



Solid state

Storage

Magnetic secondary storage devices/media

Hard drive

How it works: A magnetic disk has a stack of magnetic platters that rotate. Moving read/write heads hover just above the surfaces and can move in and out over the disc surfaces to read or write the data.

Uses: Permanent storage for a computer. Stores the operating system, program files and personal data.

Capacity: Can be a number of terabytes in size, for example 2 TB.

Speed: Fairly quick, although the discs need to spin and the read/write heads need to move in and out to find the data.

Reliability and durability: Hard drives can be damaged by impact. The heads can be damaged by impacting with the surface, which can also damage the discs.

Portability: You can get portable magnetic hard drives, but you run the risk of them getting damaged. They are most commonly found inside a computer to protect them from damage.

Cost: The cost per gigabyte is very low. A 2 TB hard drive costs about £60, which works out at about 3p per gigabyte!



Tape drive/data tape

How it works: A tape drive/data tape works like a cassette drive/cassette tape. Motors are used to wind the tape and a read/write head is used to read data from or write data to the tape.

Uses: To back up or archive data.

Capacity: Can be a number of terabytes in size, for example 6 TB.

Speed: Very slow, because the data has to be read sequentially. This means that, if the data you need is at the end of the tape, the whole tape has to be read before the data you want is reached.

Reliability and durability: Very reliable. The tape is well protected and the tapes are not easily damaged.

Portability: The tape is very portable, although it isn't usually moved around because data is backed up to data tape and then it is stored to keep it safe.

Cost: The cost of data tape per gigabyte is very low. A 6 TB tape will cost you about £40; that is less than 1p per gigabyte!



Storage

Optical secondary storage devices/media

CD drive/CD

How it works: A light laser is used to read the data on the disc. The disc is spun at speed to allow the laser to read the data.

Uses: CDs were primarily used to store music, but they can also be used to store files.

Capacity: Low, a CD stores only 700 MB.

Speed: Relatively slow because the disc has to spin and it spins at a lower rate than the disc in a hard drive.

Reliability and durability: CDs can be easily scratched or broken if not cared for properly.

Portability: CDs are extremely portable. They are thin and can be easily posted or attached to a magazine.

Cost: CDs are very cheap, normally a few pence each.



DVD drive/DVD

How it works: A light laser is used to read the data on the disc. The disc is spun at speed to allow the laser to read the data.

Uses: DVDs are primarily used to store films, but they can also be used to store files.

Capacity: Relatively low: 4.7 GB for a single-layered disc and 8.5 GB for a dual-layered disc.

Speed: Relatively slow because the disc has to spin and it spins at a lower rate than the disc in a hard drive.

Reliability and durability: DVDs can be easily scratched or broken if not cared for properly.

Portability: DVDs are extremely portable. They are thin and can be easily posted or attached to a magazine.

Cost: Relatively cheap; a DVD normally costs less than £1.



Storage

Blu-ray Disc

How it works: A light laser is used to read the data on the disc. The disc is spun at speed to allow the laser to read the data.

Uses: Blu-ray Discs are primarily used to store high-definition films, but they can also be used to store files.

Capacity: Average: 25 GB for a single-layered disc and 50 GB for a dual-layered disc.

Speed: Relatively slow because the disc has to spin and it spins at a lower rate than the disc in a hard drive.

Reliability and durability: Blu-ray Discs can be easily scratched or broken if not cared for properly.

Portability: Blu-ray Discs are extremely portable. They are thin and can be easily posted or attached to a magazine.

Cost: Cheap; a Blu-ray Disc normally costs a couple of pounds.



Storage

Solid State secondary storage devices/media

Memory stick

How it works: A memory stick is flash memory, with memory chips attached to a circuit board. It has no moving parts and makes no noise. Most memory sticks have a USB connector on the end so they can be plugged into a computer.



Uses: Storing personal data and moving it between locations.

Capacity: Capacity varies, although most memory sticks store between 32 GB and 128 GB of data.

Speed: Very quick because they are flash memory and this provides very fast retrieval. The limiting factor is the speed at which the data can be transferred over USB to the computer.

Reliability and durability: Very reliable. All flash memory has a maximum number of times that data can be read from or written to the device, but this number is very large.

Portability: Memory sticks are compact and lightweight and this makes them very portable.

Cost: The cost per gigabyte is fairly cheap. A 128 GB memory stick costs about £35, which works out at about £0.27 per gigabyte.

Memory card

How it works: A memory card is flash memory, with memory chips attached to a circuit board. It has no moving parts and makes no noise. Memory cards plug into other devices.

Uses: Used in many different devices, including cameras to store photographs and smartphones to store apps and data. They can also be used for personal storage.



Capacity: Capacity varies, although most memory cards store between 16 GB and 128 GB of data.

Speed: Very quick because they use flash memory and this provides very fast retrieval.

Reliability and durability: Very reliable. All flash memory has a maximum number of times that data can be read from or written to the device, but this number is very large.

Portability: Memory cards are compact and lightweight and this makes them very portable.

Cost: The cost per gigabyte is fairly cheap. A 128 GB memory card costs about £30, which works out at £0.23 per gigabyte.

Storage

Solid State Drive

How it works: A Solid State Drive (SSD) is flash memory, which memory chips attached to a circuit board. It has no moving parts and makes no noise. Solid State Drives can be internal to a computer or external, connecting to a computer with a USB connector. Solid State Drives use a lot less power than magnetic hard drives because as they don't need power to spin discs.



Uses: Solid State Drives are replacing magnetic hard drives in most computers because their speed. Therefore, they are commonly used to store the operating system, program files and personal data.

Capacity: Capacity varies and is increasing over time. You can currently get Solid State Drives that hold 1 TB of data.

Speed: Extremely quick because they use flash memory and this provides very fast retrieval. Internal SSDs are quicker than external SSDs because they connect directly to the motherboard.

Reliability and durability: Very reliable. All flash memory has a maximum number of times that data can be read from or written to the device, but this number is very large.

Portability: Solid State Drives are compact and lightweight and this makes them a very good option for laptops as well as desktop computers.

Cost: The cost of SSDs is coming down but they are still more expensive than magnetic hard discs. A 1TB drive will cost you about £100, which works out at about £0.10 per gigabyte.

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SHOP

Desktop computers

21.5" All-in-One PC Hachette Dual Core, 128 GB SSD, Black

Cost: £349.00

Processor: AMD Athlon Silver 3050U, Dual Core, 2.3 GHz
RAM: 4 GB
Storage: 128 GB SSD
Screen Size: 21.5"
Screen Resolution: 1920 x 1080
Graphics Card: Integrated



HodTech 7 Desktop PC – 2 TB HDD & 512 GB SSD, Black

Cost: £1500.00

Processor: Intel® Core™ i7-9700 Processor, Octa Core, 3.9 GHz
RAM: 16 GB
Storage: 2 TB Hard Drive, 512 GB SSD
Screen Size: 27"
Screen Resolution: 3840 x 2160
Graphics Card: Dedicated Graphics, NVIDIA GeForce RTX 2060



HodTech 5, 1 TB HDD & 256 SSD, Black

Cost: £699.00

Processor: Intel® Core™ i5-10400 Processor, Hexa Core, 2.9 GHz
RAM: 8 GB
Storage: 1 TB Hard Drive 256 GB SSD
Screen Size: 24.0"
Screen Resolution: 1920 x 1080
Graphics Card: Integrated



Desktop computers

Hachette Octa 5, GTX 1650, 1 TB HDD & 256 GB SSD

Cost: £1099.00

Processor: Intel® Core™ i7-10750H

Processor, Octa Core, 2.0 GHz

RAM: 8 GB

Storage: 1T B SSD

Screen Size: 27"

Screen Resolution: 1920 x 1080

Graphics Card: Integrated graphics



Hachette Hexa 5, GTX 1650, 1 TB HDD & 256 GB SSD

Cost: £799.00

Processor: Intel® Core™ i5-9400F

Processor, Hexa Core, 2.9 GHz

RAM: 8 GB

Storage: 1 TB Hard Drive, 256 GB SSD

Screen Size: 24.0"

Screen Resolution: 3840 x 2160

Graphics Card: Dedicated graphics,
NVIDIA GeForce GTX 1650



21.5" All-in-One PC HodTech 3, 128 GB SSD, Black

Cost: £399.00

Processor: Intel® Core™ i3-10110U

Processor, Dual Core, 2.1 GHz

RAM: 4 GB

Storage: 128 GB SSD

Screen Size: 21.0"

Screen Resolution: 1920 x 1080

Graphics Card: Integrated graphics



Laptops

15.6" HodTech 3 Laptop – 1 TB HDD, Black

Cost: £299.97

Processor: Intel® Core™ i3-7020U
Processor, Dual Core, 2.3 GHz
RAM: 4 GB
Storage: 1 TB Hard Drive
Screen Size: 15.6"
Screen Resolution: 1920 x 1080
Graphics Card: Integrated
Battery Life: Up to 8 hours



11.6" Laptop HodTech 2.0, 32 GB eMMC, Grey

Cost: £199.00

Processor: Intel Celeron N4000
Processor, Dual Core, 1.1 GHz
RAM: 2 GB
Storage: 32 GB SSD
Screen Size: 11.6"
Screen Resolution: 1366 x 768
Graphics Card: Integrated
Battery Life: Up to 13 hours



15.6" HodTech Gamer Laptop, RTX 2060, 512 GB SSD, Camo

Cost: £1299.00

Processor: Intel® Core™ i7-10750H
Processor, Quad Core, 2.5 GHz
RAM: 16 GB
Storage: 512 GB SSD
Screen Size: 15.6"
Screen Resolution: 1920 x 1080
Graphics Card: Dedicated graphics, NVIDIA GeForce RTX 2060
Battery Life: Up to 8 hours



Laptops

14" Laptop Hachette Quad 5, 256 GB SSD, Grey

Cost: £499.00

Processor: Intel® Core™ i5-10210U
Processor, Quad Core, 1.6 GHz
RAM: 8 GB
Storage: 256 GB SSD
Screen Size: 14.0"
Screen Resolution: 1920 x 1080
Graphics Card: Integrated
Battery Life: Up to 12.5 hours



HodTech Flip 14 2 in 1 Laptop – 256 GB SSD, Grey

Cost: £849.00

Processor: AMD Ryzen 5 3500U Processor,
Quad Core, 2.1 GHz
RAM: 8 GB
Storage: 256 GB SSD
Screen Size: 14.0"
Screen Resolution: 1920 x 1080
Graphics Card: Dedicated graphics, AMD
Vega 8
Battery Life: Up to 9 hours



15.6" Hachette Dual, 1 TB HDD, Grey

Cost: £329.00

Processor: AMD Athlon Silver 3050U
Processor, Dual Core, 2.3 GHz
RAM: 4 GB
Storage: 1T B Hard Drive
Screen Size: 15.6"
Screen Resolution: 1920 x 1080
Graphics Card: Integrated graphics
Battery Life: Up to 6 hours

